

Pre-trained models

Pre-trained models are organized into distinct categories as below:

1.Computer Vision:

Pre-Trained Models for Image Classification:

Assigns a single label to an entire image.

1. ResNet (Residual Networks)

Variants: ResNet-50, ResNet-101, ResNet-152.

Applications: General image classification, object detection, and feature extraction.

2. Inception (GoogLeNet)

Variants: Inception v3, Inception v4, Inception-ResNet.

Applications: General image classification, object detection, and transfer learning.

3. VGG (Visual Geometry Group)

Variants: VGG-16, VGG-19.

Applications: General image classification and feature extraction.

4. EfficientNet

Variants: EfficientNet-B0 to EfficientNet-B7.

Applications: General image classification and transfer learning.

5. DenseNet (Dense Convolutional Network)

Variants: DenseNet-121, DenseNet-169, DenseNet-201.

Applications: General image classification and feature extraction.

6. MobileNet

Variants: MobileNetV1, MobileNetV2, MobileNetV3.

Applications: Mobile image classification and embedded vision applications.

7. NASNet (Neural Architecture Search Network)

Variants: NASNet-A, NASNet-B, NASNet-C.

Applications: General image classification and transfer learning.

8. Xception (Extreme Inception)

Applications: General image classification and transfer learning.

9. AlexNet

Applications: General image classification and historical benchmarks.

10. Vision Transformers (ViT)

Applications: General image classification and large-scale vision tasks.

Pre-Trained Models for Object detection:

1. The YOLO Series (You Only Look Once)

Best For: Real-time applications, edge devices, and mobile deployments.

2. Transformer-Based Detectors (DETR, RT-DETR, RF-DETR)

Best For: Complex, crowded scenes, occluded objects, and maximum cloud-based accuracy.

3. Two-Stage Detectors (Faster R-CNN)

Best For: High-accuracy projects where real-time latency is not critical.

4. Lightweight Mobile Detectors (SSD / MobileNet)

Best For: Low-power environments, older Android/iOS smartphones, and Raspberry Pi builds.

Pre-Trained Models for Image Segmentation:

1. Foundational & Zero-Shot Models:

- Segment Anything Model (SAM / SAM 2)
- YOLO (e.g., Ultralytics YOLOv8 / YOLO11)

2. Classic & Architecture-Specific Backbones

- U-Net
- DeepLabv3 / DeepLabv3+
- Mask R-CNN
- SegFormer

2.NLP:

Pre-Trained Models for Text Classification:

1. BERT (Bidirectional Encoder Representations from Transformers)
2. RoBERTa (Robustly Optimized BERT Approach)
3. DistilBERT:
4. FastText
5. T5 (Text-to-Text Transfer Transformer)
6. GPT Series (Generative Pre-trained Transformer)
7. BART / XLNet:
8. GPT-2 (Generative Pretrained Transformer 2)
9. ELMo (Embeddings from Language Models)
10. Transformer-XL

- 11.Binary-Partitioning Transformer (BPT)
- 12.Neural Attentive Bag-of-Entities Model for Text Classification (NABoE)
- 13.Rethinking Complex Neural Network Architectures for Document Classification

Pre-Trained Models for Text generation and completion:

1. GPT-4o / GPT-4
2. Llama 3 / 3.1
3. Mistral / Mixtral
4. Claude 3.5 Sonnet
5. Qwen

3.Audio and speech processing:

Pre-Trained Models for Speech-to-Text (ASR): Whisper

Pre-Trained Models for Text-to-Speech (TTS): Bark

Pre-Trained Models for Speech Enhancement: Demucs

Pre-Trained Models for Audio Classification: Audio Spectrogram
Transformer (AST)

Pre-Trained Models for Speech Feature Extraction: Wav2Vec 2.0

Pre-Trained Models for Language Speech Tasks: MMS

4.Multimodal(Cross Modal):

Pre-Trained Models for Vision-Language Alignment: CLIP (OpenAI) and ALIGN (Google)

Pre-Trained Models for Multimodal LLMs (VLM): GPT-4o, Claude 3.5 Sonnet, and Gemini 1.5 Pro

Pre-Trained Models for Open-Weights VLMs: LLaVA (Large Language and Vision Assistant) and PaliGemma (Google)

Pre-Trained Models for Text-to-Image / Video: Stable Diffusion (Stability AI), Midjourney, and Sora (OpenAI)